1. High-Throughput Speculative Decoding LLM Engine

- **Reference GitHub Projects:** vllm-project/vllm & vllm-project/speculators
- **The Concept:** Standard LLM serving is memory-bandwidth bound. Projects like vLLM use PagedAttention and Speculative Decoding (using a tiny draft model like Llama-3.2-1B to propose tokens and a larger model to verify them in parallel).
- **Your Adaptation:** Build a custom lightweight Speculative Inference Engine in C++/Python that implements:
  - Dynamic batching of candidate token trees (e.g., using Medusa or EAGLE-style draft heads).
  - Paged KV-Cache allocation to eliminate VRAM fragmentation during parallel verification passes.
- **Resume Impact:** *"Built a C++/PyTorch speculative decoding engine based on vLLM patterns, optimizing token draft verification to achieve a 2.1x speedup in TTFT (Time-To-First-Token) for local 8B LLMs."*

2. Embedded Graph-Vector Hybrid Search Engine

- **Reference GitHub Projects:** HelixDB/helix-db & qdrant/qdrant
- **The Concept:** Modern AI applications require both vector similarity (for semantic search) and graph structures (for relational context). HelixDB and Qdrant tackle this by fusing graph traversal with HNSW vector indexes in Rust.
- **Your Adaptation:** Engineer a low-latency embedded database in Rust or Go that combines:
  - **HNSW (Hierarchical Navigable Small World)** graph indexing for high-dimensional vector search.
  - **Property Graph storage** with ACID-compliant memory-mapped IO (mmap).
  - A unified query execution planner that filters vector candidate sets using graph node adjacency constraints.
- **Resume Impact:** *"Engineered a hybrid Graph-Vector database in Rust using HNSW indexing and mmap persistence, enabling sub-15ms joint graph-traversal vector queries across 500k+ embedding nodes."*

3. Privacy-First Local Context & Screen Indexing Engine

- **Reference GitHub Projects:** mediar-ai/screenpipe & Logseq
- **The Concept:** Screenpipe is a Rust-based, 100% local background engine that records audio/screen context, runs local OCR and Whisper transcriptions, and indexes everything into SQLite for contextual AI querying.
- **Your Adaptation:** Construct a local ambient context pipeline that continuously captures system audio and active window frames, featuring:
  - On-device OCR (using Tesseract or Apple Vision API) and quantized Whisper STT execution.
  - Sliding-window vector embeddings written to a local SQLite/Vec extension.

    - A local RAG agent that answers questions about what you worked on, read, or discussed during the day without cloud telemetry.
- **Resume Impact:** *"Architected a 100% local ambient context capture engine in Rust & SQLite that indexes screen text and audio streams in real time, serving sub-second semantic search over historical personal workflows."*

4. Active-Active Local-First Sync Engine

- **Reference GitHub Projects:** electric-sql/electric & automerge/automerge
- **The Concept:** ElectricSQL and Automerge are leading the shift to "local-first" software. They stream Postgres changes directly to client-side local databases (like SQLite/PGLite) using Conflict-free Replicated Data Types (CRDTs) to handle offline edits seamlessly.
- **Your Adaptation:** Build a bi-directional replication daemon between a central database and client-side SQLite stores:
    - Implement logical replication slot listeners (e.g., Postgres pgoutput) to capture change streams.
    - Use operation-based CRDTs (e.g., LWW-Element-Set) for deterministic multi-client conflict resolution.
    - Package a lightweight TypeScript/Rust client library that handles offline optimistic mutations and background resynchronization.
- **Resume Impact:** *"Designed an active-active sync daemon implementing CRDT conflict resolution over Postgres WAL streams, allowing offline client mutations to converge deterministically upon reconnection."*

Comparison Matrix for Selection

Project Concept	Primary Stack	Architecture Level	Key Research / Spec Area
Speculative Inference Engine	C++, PyTorch, CUDA	System & AI Infra	Memory bandwidth optimization & KV-cache management
Graph-Vector Database	Rust / Go, HNSW	Storage & Database Internals	High-dimensional geometric index structures
Local Ambient Context Engine	Rust, SQLite, ONNX/Whisper	Edge Computing & Multimodal AI	Real-time signal processing & local vector indexing
Local-First Sync Engine	TypeScript, Elixir/Rust, SQLite/Postgres	Distributed Systems & Networks	Logical replication, state convergence, & CRDTs